

Chapter11:WEB MULTIMEDIA AND INTERACTIVITY

Containers & Codecs

Container

- Designated by the file extension – contains the media and metadata

Codec

- The algorithm used to compress the media HTML5 audio & video
- Native to the browser
- ISSUE: Browsers do not all support the same codecs
 - <http://www.longtailvideo.com/html5/> (2017)
 - <https://caniuse.com/#search=video%20format> (current)

Common Audio File Types

- .wav Wave File
- .aiff Audio Interchange File Format
- .mid Musical Instrument Digital Interface (MIDI)
- .au Sun UNIX sound file
- .mp3 MPEG-1 Audio Layer-3
- .ogg Ogg Vorbis (open-source)
- .m4a MPEG 4 Audio.
This audio-only MPEG-4 format is supported by Quicktime, iTunes, and iPods.

Common Video File Types

.avi	Microsoft Audio Video Interleaved
.av1	Alliance for Open Media (open-source)
.wmv	Windows Media File
.mpg	MPEG (Motion Picture Experts Group)
.m4v .mp4	(MPEG-4)
.ogv	Ogg Theora (open-source)
.webm	VP8 codec (open video format, free)

Configure Audio & Video

Hyperlink

The most basic method to provide audio or video files

```
<a href="wdfpodcast.mp3" title="Web Design Podcast">Web Design Podcast</a>
```

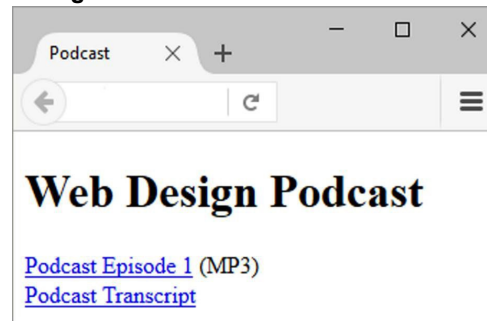


Figure 11.1 The default MP3 player will launch in the browser when the visitor clicks on Podcast Episode 1

HTML5 Audio & Source Elements

```
<audio controls="controls">  
    <source src="soundloop.mp3" type="audio/mpeg">  
    <source src="soundloop.ogg" type="audio/ogg">  
    <a href="soundloop.mp3">Download the Audio File</a> (MP3)  
</audio>
```

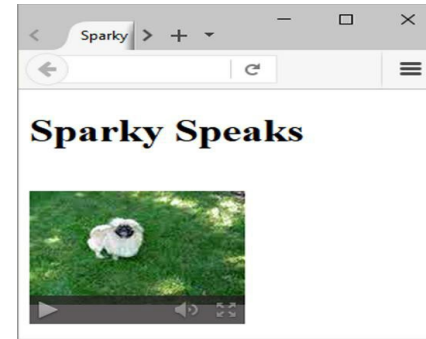


Figure 11.4 Audio & Source Elements

HTML5 Video & Source Elements

```
<video controls="controls" poster="sparky.jpg" width="160" height="150">  
    <source src="sparky.m4v" type="video/mp4">  
    <source src="sparky.ogv" type="video/ogg">  
    <a href="sparky.mov">Sparky the Dog</a>  
(.mov)  
</video>
```

Copyright Issues

- Only publish web pages, images, and other media that you have personally created or have obtained the rights or license to use.
- Ask permission to use media created by another person instead of simply “grabbing” it.
- All work (including web pages) are automatically copyrighted even if there is not copyright mark or date.
- Fair Use Clause of the Copyright Act
- Creative Commons – A new approach to copyright

Revision of JavaScript

- Object-based client-side scripting language
- Originally developed by Brendan Eich at Netscape
- Manipulates the objects associated with a web page document:
 - the window
 - the document
 - the elements such as forms, images, hyperlinks, and so on

JavaScript Examples

Example 1: The use of the alert box and internal documentation (comments) in JavaScript.

```
<html>
<head>
<title>2057 JavaScript</title>
<script language="JavaScript" type="text/javascript">
/* multi-line comment,
this is a huge block of
text that I've commented
out */
// single line comment - put up an alert box, to show how they work
alert("Hello to your first JavaScript");
</script>
</head>
<body>
</body>
</html>
```

Example 2: The use of multiple alert boxes in JavaScript.

```
<html>
<head>
<title>2057 JavaScript</title>
<script language="JavaScript" type="text/javascript">
// put up an alert boxes, to show how they work
alert("Here's the first one");
alert("The second one");
alert("And the third one");
</script>
</head>
<body>
<h1>Yes, I have seen these and now I know it!</h1>
</body>
</html>
```

Example 3: The use of link events in JavaScript and internal documentation(comments) in HTML.

```
<html>
<head>
<title>2057 JavaScript</title>
<script language="JavaScript" type="text/javascript">
</head>
<body>
<h1>Link Events Example</h1>

<!-- This is a link event -->
<!-- When someone clicks on this link, run the little bit of JavaScript between the quotes -->
<a href="#" onClick="alert('Try it again!');">Click on me!</a>

<!-- This is like onClick except it uses onMouseOver -->
<a href="#" onMouseOver="alert('Neat Eh!');">Mouse over me!</a>
</body>
</html>
```

Example 4: The use of image swapping in JavaScript.

```
<html>
<head>
<title>2057 JavaScript</title>
<script language="JavaScript" type="text/javascript">
</head>
<body>
<h1>Image Swapping Example</h1>

<a href="#" onMouseOver="document.the_image.src='smallibm.gif';"
onMouseOut="document.the_image.src='swoosh.gif';">

</a>

</body>
</html>
```

NOTE: You will need two images that are the same size (height and width) and in your Web page directory. The example above uses the images named 'smallibm.gif' and 'swoosh.gif'.

Summary

This chapter introduced many HTML used to configure sound, video, and interactivity on web pages.